Fidan Karimli

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EDUCATION

BSc in Process Automation Engineering, IT Department, Baku Higher Oil School 09/ 2021 – 06/ 2026
- Presidential Scholarship(being 44th through 50000 students), funded by the Government of Azerbaijan.

INTERNATIONAL EXPERIENCE

Petnica Summer Institute of Machine Learning (PSI:ML), Serbia Aug 2026
- Selective residential research program in scientific ML (12.5% acceptance); scholarship, sole participant from Azerbaijan. Built a self-supervised GNN (JEPA pretraining on 6,782 structures, fine-tuned on 271,231 MegaScale mutations) to predict single-mutation protein destabilization. Link for the project.

"Future Energy Leaders" Summer School, CUP, Beijing, China 06/2025 – 07/ 2025
- Awarded a fully funded scholarship as one of two selected representatives from Azerbaijan
- Led a 5-member international team and worked with peers from 16 countries on a sustainable energy project, promoting teamwork and cross-cultural collaboration.

Multiculturalism Exchange Program, Universidad de León, Spain 10/ 2023 – 12/ 2023
- Awarded a fully funded scholarship for the GearUp! Exchange Programme.
- Developed a global network of peers from various cultures, fostering ongoing collaboration and shared learning.

RESEARCH EXPERIENCE & PUBLICATIONS

- **Research Collaborator, Istanbul Technical University** 2026–present
  Collaborating with Prof. Yusuf Yaslan's group on cross-market recommendation systems using GNN-based approaches and the XMarket dataset, focusing on fairness for non-overlapping users.

- A. Najafli, K. Oghuz, **F. Karimli**, Y. Mammadli, R. Khidirov, E. Abdullayev. "Intelligent Structural Hazard Detection and Adaptive Hydraulic Response for Underground Coal Mines." In *Proc. 20th IEEE International Conference on Application of Information and Communication Technologies (AICT)*, Baku, Azerbaijan, Oct. 2026. - **Accepted**

PROFESSIONAL EXPERIENCE

Junior AI Engineer, National AI Center of Azerbaijan 08/2025 – present
- Adapting open-weight LLMs (Qwen2.5, Qwen3) to Azerbaijani legal text via SFT and LoRA, including multi-node fine-tuning on an H200 GPU cluster.
- Extending adaptation across modalities: fine-tuning vision-language models for OCR and structured extraction from Azerbaijani court documents.
- Applying graph neural networks to legal judgment prediction, modelling cases, statutes, and procedural relations as graph structure.
- Built the retrieval stack behind LEGALENS: dense retrieval over 1.6M documents (Vespa) with cross-encoder re-ranking; applications conducted a literature survey of legal IR methods,(SAILER, LegalDuet, DELTA, SEMDR).

Note: (Intern Apr–Jul 2025)

Data Analyst, AGA Group 05/2024 – 04/2025
- Transformed raw data into actionable insights for finance and marketing teams using SQL, Power BI, and Python.
- Applied Pareto (80/20) analysis to identify key sales drivers and developed corresponding KPIs.

Data Science Mentor, Lotfizadeh Technology Center 05/2024 – 02/2025
- Mentored students on **regression and classification algorithms**, guiding them through coursework and final projects.

Automation Engineer Intern, Emerson 06/2023 – 07/2023

- Worked on valve automation projects, including **testing, calibration, and performance optimization.**
- Supported the engineering team in developing efficient and reliable automation solutions.

ACTIVITIES & LEADERSHIP

Azerbaijan Ambassador, WiDS (Women in Data Science Worldwide) 11/2024 – present

- Advocating for gender diversity and fostering an inclusive environment in **STEM fields**.
- Organizing impactful events, workshops, and networking opportunities to inspire women in tech and data science.

President of Data Science Club, Baku Higher Oil School 01/2024 – 04/2026

- Led the planning and execution of workshops, hackathons, and training sessions to develop members' technical skills.
- Mentored younger club members, supporting their personal and professional growth, much like how I would mentor youth leaders for peace initiatives.

Volunteer, Azerbaijan Red Crescent Society 06/2023 – 08/2024

- Coordinated donation drives and distributed essential supplies to those in need.

TECHNICAL SKILLS

- **Programming:** Python, SQL, MATLAB, Git
- **Machine Learning:** PyTorch, HuggingFace Transformers, PEFT/LoRA, Graph Neural Networks, Neural ODEs, Recommender Systems, Scikit-Learn
- **NLP & Information Retrieval:** Dense Retrieval (Vespa), Cross-encoder Re-ranking, RAG, Sentence Transformers
- **Data & Tools:** NumPy, Pandas, Matplotlib, Seaborn, Plotly, Docker,

AWARDS & HONORS

- **'Best Presentation" Award by Teknofest 2025** 2025
- **1st place at Maire&BHOS Energy Transition Competition winner** 2025
- **European Union Green Business Case Competition 2nd place winner** 2024
- **Successful Paper Presentation –** International Scientific Conference on "Sustainable Environment" 2024
- **1st place –** International "TransitHack" Hackathon by Huawei and Socar 2024
- **2nd place –** Sustainability Ideation Challenge by Innovation and Digital Development Agency of Azerbaijan
- **McKinsey & Company Forward Program** 2024
- **Honor Degree –** successful completion of a Data Science Course at LTC Lab (Grade A) 2023
- **Winner of the "IRON CODE" Scholarship** by AGA Group 2023
- **Presidential Scholar –** Top 25 out of 67,000 Students Nationwide 2021
- **Red Diploma –** Completing high school with an overall score of 99 2021
- **3rd place at National Olympiad in Mathematics –** 2015
  Awarded by the Ministry of Science and Education of Azerbaijan

LANGUAGES

- Azerbaijan – Native Language
- English – *IELTS 7.0 out of 9.0*
- *Turkish – Full Working Proficiency*